

Barangay Health Center System - Workflow & Process Summary

This document summarizes how the BHC System works end to end: who does what, how patients move through the health center, and how data flows through the application. For step-by-step instructions and URLs, see [USER_GUIDE.md](#).

1. Purpose

The system combines a **patient registry** with a **multi-station queue** so barangay health centers can:

- Keep one record per patient (BHC ID)
 - Route each visit to the right service (Triage, Consultation, Pharmacy)
 - Serve patients in fair order with visible "Now serving" numbers
 - Alert waiting patients via **display screen** and **phone ticket** (flash + optional chime)
 - Reduce paper tickets using QR codes on patient phones
 - Manage **staff logins** (admin) and track activity / monthly statistics for supervisors
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2. Main actors

Actor	Role in the process
Registration staff	Register or find patients; create routed queue tickets
Station staff (Triage, Consult, Pharmacy)	Call, serve, complete, skip, or recall patients at their station
Coordinator / supervisor	Monitor all stations from Queue Management ; recall skipped patients when they return
Admin	Full operations plus Monthly Reports, Activity Log, and Staff accounts
Patient	Waits with ticket number; watches display or phone ticket; no login required

3. System access (login and roles)

Staff and administrators use the same **login** page but land on **different dashboards** and menus.

Role	After sign-in	What they see
Staff	Staff dashboard (/staff)	Patient Registry, Queue Stations, Queue Management
Admin	Admin dashboard (/admin)	Everything staff has, plus Administration : Monthly Reports, Activity Log, Staff accounts

Wrong-role access: If staff open `/admin` (or other admin URLs), they are **redirected** to the staff dashboard with an error message. If an admin opens `/staff`, they are redirected to the admin dashboard with an info message. This avoids a blank "Forbidden" page.

Staff accounts (admin only) — `/users`:

- Add users with role `staff` or `admin`; reset passwords; activate or deactivate
- Cannot deactivate your own account or the **last active admin**
- Logged as `user_create`, `user_deactivate`, `user_activate`, `user_password_reset`

Staff cannot open admin pages by typing the URL directly.

4. Stations and responsibilities

ID	Station	Role in workflow
1	Registration (Patient Routing)	Search patient, enter reason for visit, assign target station. Does not hold a service queue — only creates tickets for other stations.
2	Triage / Vitals	First clinical stop when routed here
3	Consultation	Doctor/nurse consultation when routed here
4	Pharmacy	Medicine dispensing; often the last stop

A single visit may use **one or more** service stations. Each new destination normally gets a **new ticket** from Registration. Exception: a **skipped** ticket at a service station can be **recalled** to waiting (same ticket number) if the patient returns the same day.

5. End-to-end process (typical visit)

Phase A — Arrival and registration

1. Patient arrives at the health center.
2. Staff opens **Patient Registry** and either:
 - **Adds** a new patient (name, sex, date of birth → system assigns **BHC ID**), or
 - **Finds** an existing patient (search / autocomplete).
3. Patient record is stored for future visits.

Phase B — Routing (Registration desk)

1. Staff opens **Patient Routing** (Registration, Station 1).
2. Staff searches patient by name or BHC ID.
3. Staff records **reason for visit** and selects **target station** (Triage, Consultation, or Pharmacy).
4. System creates a **queue ticket** for that station:
 - Status: `waiting`
 - Ticket number: station prefix + daily sequence (e.g. `TR-001`, `CN-002`, `PH-003`)
5. Staff shows the patient the **ticket screen** (QR). Patient may scan to follow status on their phone.

Phase C — Waiting and patient alerts

1. Patient waits in the area for the assigned station.
2. **Waiting area display** (/display/{stationId}) — for TV/tablet in the waiting room:
 - Shows **Now serving** and **Next** waiting tickets
 - Refreshes every **3 seconds**
 - **Flashes** when Now serving changes
 - **Chime** when Now serving changes, after staff tap **Enable sound** once (browser requirement)
3. **Patient ticket page** (/ticket/{ticketId}) — on the patient's phone:
 - Refreshes every **5 seconds**
 - When status becomes **SERVING**: message "You are being called now"
 - **Chime** on that transition, after patient taps **Enable sound** once
4. Staff may still **call names verbally**; display and ticket are digital backups.

The system does **not** send SMS or mobile push notifications. Patients rely on the TV display, the QR ticket page, and staff voice.

Phase D — Service at station

1. Station staff opens that station's queue (**Queue Stations**) or uses **Queue Management** for all stations.
2. Staff actions:

Action	Effect
Call next	Oldest waiting ticket → serving (Now serving)
Call (specific ticket)	That ticket → serving
Complete	Ticket → done ; step finished at this station
Skip	Ticket → skipped (e.g. patient absent, did not hear call)
Recall to queue	Today's skipped ticket at this station → waiting again (same ticket number)

3. **Auto-call**: After **Complete** or **Skip**, if no one is **serving** and more patients are **waiting**, the system **automatically calls the next** patient (ticket_call_next_auto).
4. **Recall does not auto-call** — staff use **Call** or **Call next** when the returned patient should be served.
5. Only **one** patient can be **serving** per station at a time.

When to recall vs new ticket

Situation	What to do
Skipped by mistake; patient still on site, same day, same station	Recall to queue on station page or Queue Management
Patient left and came back later, or different station needed	New ticket from Registration (routing)
Recall not allowed (wrong day or not skipped)	New ticket from Registration

Phase E — Further stations or end of visit

- If the patient needs **another service**, staff returns to **Registration** and creates a **new ticket** for the next station.
- When finished (usually after Pharmacy), staff marks **Complete** at the final station.
- Tickets remain in the database for **monthly reports** and history (skip and recall events stay in the **Activity Log**).

6. Ticket lifecycle

See [WORKFLOW_DIAGRAM.md](#) (ticket state diagram) and [ticket-lifecycle.mmd](#) for a visual of ticket statuses.

Status	Meaning
waiting	In line, not yet called
serving	Currently being served (Now serving)
done	Successfully completed at this station
skipped	Not served (absent, etc.); may recall to waiting same day at same station

Queues are **per station, per calendar day** (`DATE(created_at) = today`).

7. Staff operating modes

Per-station mode

Queue Stations → open station → **Call** / **Complete** / **Skip** / **Recall** (on skipped rows in **Today's tickets**). Best when one staff member owns one desk.

Queue Management mode (busy days)

One screen (`/coordinator`) shows **all service stations** (except Registration):

- Waiting and serving counts per station
- **Now serving** with Complete / Skip
- **Call next** and per-ticket **Call** for the next waiting list
- **Skipped today** with **Recall** (same rules as station page)
- Auto-refresh every **5 seconds**
- Same **auto-call** rules after Complete / Skip

Use when one person oversees multiple lines or during peak hours.

8. Patient notification flow

Channel	How it works
TV / monitor	Fullscreen <code>/display/{stationId}</code> ; Enable sound once; flash + chime when Now serving changes; refresh every 3 s

Channel	How it works
Phone (QR ticket)	<code>/ticket/{ticketId}</code> ; Enable sound once; chime when status becomes SERVING ; refresh every 5 s
Staff queue actions	Call / Call next update the database; they do not play sound on the staff PC — alerts run on display/ticket only
Verbal	Staff call names; digital channels supplement

LAN note: For QR codes on phones, `base_url` in `config/app.php` must use the server's **LAN IP**, not `localhost`.

Browser note: Chrome and tablets block audio until the user taps **Enable sound** (or interacts with the page). Open the display each morning and enable sound once.

9. Administration and oversight

Staff accounts (admin)

- Managed at `/users` (sidebar: **Staff accounts**)
- Onboard registration and station staff without database tools
- Default `admin` / `admin123` on first install if no users exist — change password immediately

Monthly Reports (admin)

- Filter by month (YYYY-MM)
- Totals: tickets created, completed, **skipped**
- Per-station breakdown and average wait/service times
- Based on ticket **creation date** in that month (recall does not remove a prior skip from counts)

Activity Log (admin)

Records accountability events, including:

- `login` / `logout` / `login_failed`
- `patient_create`, `patient_update`
- `ticket_create`, `ticket_call`, `ticket_call_next`, `ticket_call_next_auto`
- `ticket_complete`, `ticket_skip`, `ticket_recall`
- `user_create`, `user_deactivate`, `user_activate`, `user_password_reset`, `user_password_change` (self-service)

10. Planned enhancements (coming soon)

These are **not** in the current build; they are listed on the **admin dashboard** for transparency:

Feature	Status
Station setup UI (add/rename stations)	To follow

Feature	Status
SMS / push reminders	To follow
Patient self check-in kiosk	To follow

11. System boundaries (what the system does not do)

- **No automatic routing between stations** — staff create a new ticket at Registration for each new destination (unless **recalling** a same-day skipped ticket at the **same** station).
- **No patient self-check-in** — all tickets are staff-created.
- **No SMS/push** — patients use display, ticket page, and staff voice.
- **No voice announcement (TTS)** — chime only; staff call names.
- **No auto-skip** for no-shows — staff must press Skip.
- **No un-skip across days** — recall only works for **today's** skipped tickets at that station.
- Registration (Station 1) does not use Call / Complete / Skip / Recall for a service queue.
- **No self-registration for staff** — only an admin creates logins under Staff accounts.

12. Physical setup (recommended)

Location	System use
Registration desk	Patient Registry + Patient Routing
Each service room	Station queue page
Optional central desk	Queue Management
Waiting areas	Display URL per station; Enable sound ; fullscreen
Patient phones	QR ticket page; Enable sound if using chime
Server PC	XAMPP or hosted PHP, MySQL, static LAN IP (or InfinityFree for cloud)

13. Workflow diagram

End-to-end process overview (patient registry, routing, waiting-area alerts, station queue, recall, and administration):

See [WORKFLOW_DIAGRAM.md](#) and [workflow-diagram.mmd](#) to preview or export the full workflow diagram.

14. Quick comparison: documents in this project

Document	Audience	Content
README.md	All	Documentation index
WORKFLOW_SUMMARY.md (this file)	Supervisors, trainers	Process overview, roles, lifecycle, recall, alerts

Document	Audience	Content
USER_GUIDE.md	Daily staff, doctors, admins	How-to, URLs, displays, clinical workflow, troubleshooting
INTEGRATED_USER_GUIDE.md	Barangay staff	Gawad BIS ↔ BHC integration
WORKFLOW_DIAGRAM.md	All	Mermaid diagrams
SOURCE_CODE_AND_DOCUMENTATION.md	Developers	Technical reference

Summary version: BHC System with staff/admin dashboards, Staff accounts, role redirects, Patient Routing, multi-station queues, QR tickets, display and ticket sound alerts, Queue Management, recall to queue, auto-call on Complete/Skip, Monthly Reports, and Activity Log.